

15.1 The Gastrointestinal Tract

There is some truth to the saying, "You are what you eat". Rest assured, however, that no matter how many hamburgers you eat, you will never become one. Instead, the hamburger becomes you. This transformation is largely due to the activities of the digestive system.

Like an assembly line in reverse, the digestive system takes the food we eat and breaks the complex organic molecules into their chemical subunits. The subunits are molecules small enough to be absorbed into the bloodstream and delivered to body cells, where they provide either materials for growth and repair of the body or energy for daily activities. Imagine that you just ate a cheeseburger on a bun. The starch in the bun may fuel a jump for joy; the protein in the beef may be used to build muscle; and the fat in the cheese may become myelin sheaths that insulate nerve fibers. The digestive system breaks food into molecules small enough to be absorbed and delivered to the cells that use them. Table 15.1 identifies the structures of the digestive system and describes their roles in both mechanical and chemical digestion. You can refer to the table as you read about each of the following digestive structures.

Table 15.1 Review of Structures of the Digestive System

Structure	Description/Functions	Mechanical Digestion	Chemical Digestion
Mouth	Receives food; contains teeth and tongue: tongue manipulates food and monitors quality	Teeth tear and crush food into smaller pieces	Digestion of carbohydrates begins
Pharynx	Passageway for both food and air	None	None
Esophagus	Tube that transports food from mouth to stomach	None	None
Stomach	J-shaped muscular sac for food storage	Churning of stomach mixes food with gastric juice, creating liquid chyme	Protein digestion begins
Small Intestine	Long tube where digestion is completed and nutrients are absorbed	Segmental contractions mix food with intestinal enzymes, pancreatic enzymes, and bile	Carbohydrate, protein, and fat digestion is completed
Large Intestine	Final tubular region of GI tract: absorbs water and ions; houses bacteria; forms and expels feces	None	Some digestion is carried out by bacteria
Anus	Terminal outlet of digestive tract	None	None

The digestive system consists of a long, hollow tube, called the gastrointestinal (GI) tract, into which various accessory glands release their secretions. The GI tract begins at the mouth and continues to the pharynx, esophagus, stomach, small intestine, and large intestine. The hollow

area of the tube through which food and fluids travel is the lumen. Along most of its length, the walls of the GI tract have four basic layers.

- **Mucosa.** The innermost layer of the GI tract is the moist, mucus-secreting layer called the mucosa. The mucus helps lubricate the lumen, allowing food to slide through easily. Mucus also helps protect the cells lining the GI tract from rough substances in the food and from digestive enzymes. In some regions of the digestive system, cells in the mucosa also secrete digestive enzymes. In addition, the mucosa in some parts of the digestive tract is highly folded, which increases the surface area for absorption.
- **Submucosa.** The next layer, the submucosa, consists of connective tissue containing blood vessels, lymph vessels, and nerves. The blood supply maintains the cells of the digestive system and, in some regions, picks up and transports the products of digestion. The nerves are important in coordinating the contractions of the next layer.
- **Muscularis.** The next layer, the muscularis, is responsible for movement of materials through the GI tract and for mixing ingested materials with digestive secretions. In most sections of the GI tract, the muscularis is a double layer of smooth muscle. (The stomach, as you will read later, is an exception; it has three layers of muscle.) In the inner, "circular" layer of muscle, the muscle cells encircle the tube, causing a constriction when they contract. In the outer, longitudinal layer, the muscle cells are arranged parallel to the GI tract. Longitudinal muscles shorten the GI tract when they contract. The muscle layers churn the food until it is liquefied, mix the resulting liquid with enzymes, and propel the food along the GI tract in a process called peristalsis, which is discussed later in this chapter. Local muscle contractions in random regions of the GI tract, especially in the small intestine, are called segmental contractions. Segmental contractions mix intestinal contents—food and digestive juices (fluids containing enzymes and other fluids that aid in digestion). Segmental contractions also help the body absorb digested food by moving intestinal contents over the intestinal wall.
- **Serosa.** The serosa, a thin layer of epithelial tissue supported by connective tissue, wraps around the GI tract. It secretes a fluid that lubricates the outside of the GI tract to reduce friction with surfaces of the intestine and other abdominal organs that come into contact with the GI tract.